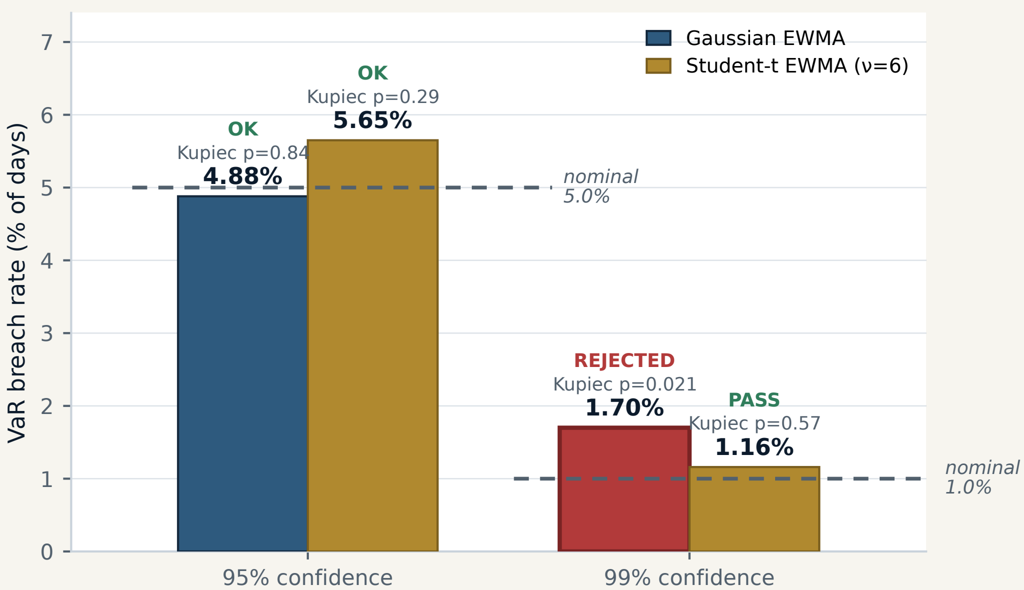
VALIDATION

VaR & ES calibration on real data

Coverage-tested against realized P&L · 1,292 trading days × 6 shipping equities (5y) · 60-day EWMA window

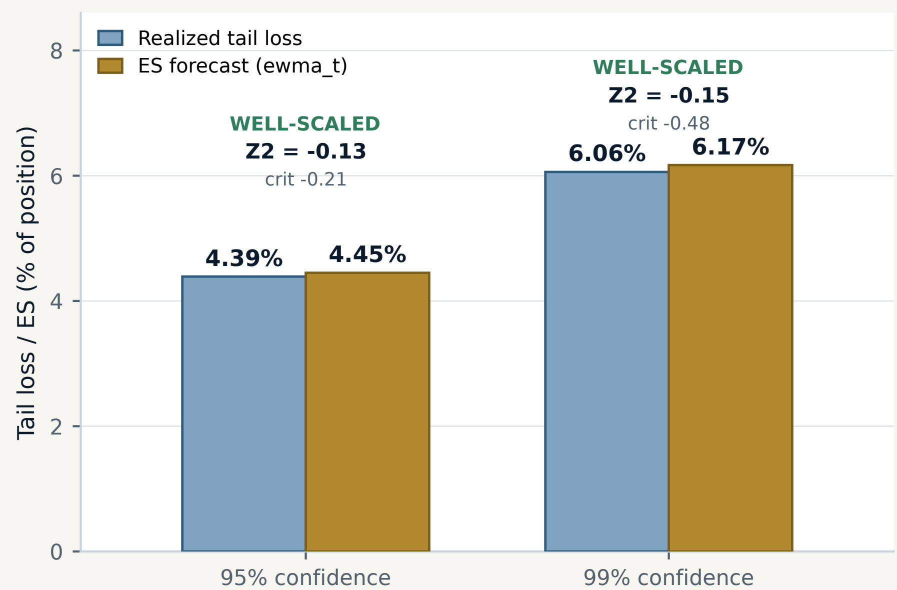
VaR coverage — breach rate vs nominal

Gaussian under-states the 99% tail (red); Student-t keeps both levels un-rejected



ES coverage — realized vs forecast (Acerbi-Szekely)

Realized tail loss tracks the Student-t ES at both levels — the ES is honest



HOW TO READ THIS

- BREACH RATE = share of days realized loss exceeded the VaR; it should sit at nominal (5.0% at 95%, 1.0% at 99%).
- KUPIEC POF: $p < 0.05$ = breach count too far from nominal to be chance → VaR REJECTED. Gaussian 99% fails ($p=0.021$); Student-t passes.
- ACERBI–SZEKELY Z2 grades the Expected Shortfall on breach days: ~0 = well-scaled; negative past the critical value = ES under-states the tail.
- Both ES levels clear their critical values, so the Expected Shortfall is honest — not just the VaR quantile.
- TAKEAWAY: the Gaussian tail under-states the 99% fat tail; Student-t (ν=6) fixes the VaR AND keeps the ES well-scaled at both confidences.
- 2 of the 20 backtest validators run on this real market cache; the other 18 are deterministic walk-forward / structural tests.